



Technical Service Bulletin: TSB150027

Released Date: 02-Sep-2015

New Cylinder Liner Crevice Seal and Associated Cylinder Liner Part Numbers

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- G19
- K19
- KTA19GC CM558/CM2358 SLB
- QSK19 CM500
- QSK19 CM850 MCRC
- QSK19 CM2150 MCRC
- QSK19 CM2350 K105
- QSK19 G CM558/CM2358
- QSK19 G CM558/CM2358 and CM700/CM2358

Description of Change

The new cylinder liner crevice seal groove now sits further down in the lower press fit area. The groove acts as a positive locator for the upper crevice seal, which minimizes vertical movement of the seal during installation. The new upper crevice seal is now a D-shape design. The seal material remains unchanged. See Figure 1 below. The previous cylinder liner crevice seal groove sat higher in the lower press fit area and did **not** have a positive locator, which could allow the seal to move vertically during installation. The previous upper crevice seal was a rectangular design. See Figure 2 below.

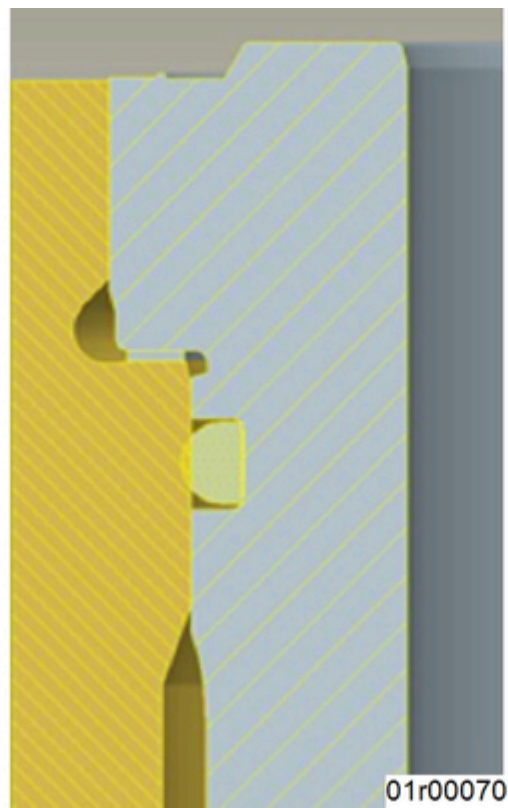


Figure 1, New Cylinder Liner and Upper Crevice Seal Installed in Cylinder Block

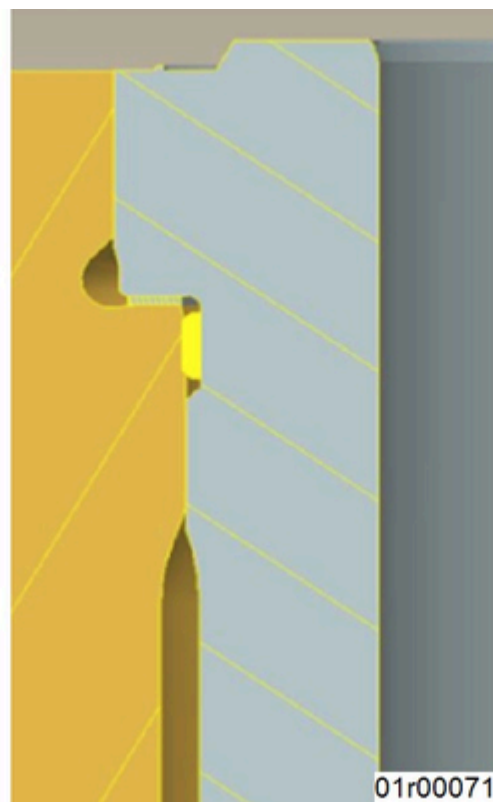


Figure 2, Existing Cylinder Liner and Upper Crevice Seal Installed in Cylinder Block

Reason for Change

The new cylinder liner and upper crevice seal is more robust and helps to reduce seal pinching between the cylinder liner flange and brass sealing ring during cylinder liner installation. This can help reduce coolant leaks from the cylinder head gasket area.

Service Parts Availability

Table 1, Upper Crevice Seal Part Number

Description	Existing Part Number	New Part Number
Upper Crevice Seal	4000504	4308855

Table 2, 3, and 4 Legend:

Definition of number identifiers listed in each column:

1 = Newly created cylinder liner size, no existing cylinder liners of this size are offered.

2 = No new cylinder liners will be released for this size. Old parts still available while stock remains. Obsoleted once stock depleted.

3 = Cylinder liner kits for new Tier 4 part numbers will be released at a later date. Prior cylinder liner kits were **not** offered.

All kits containing the individual cylinder liner part numbers below will now contain the new cylinder liner part numbers and new upper crevice seals.

The three seals on the lower portion of the cylinder liner remain unchanged.

Table 2, Individual Cylinder Liners and Corresponding Kits*

Existing Cylinder Liner Kit Part Number	Existing Cylinder Liner Part Number	New Cylinder Liner Kit Part Number	New Cylinder Liner Part Number	Upper Press Fit Oversize [inches]	Lower Press Fit Oversize [inches]	Size [inches]
4024770	4009227	4371770	4333984	-0.01	0	-0.01/0
4024767	4009220	4371769	4308809	0	0	0/0
1	1	4371774	4371641	0.01	0.01	.010/.010
4024773	4009231	4371771	4333987	0.02	0.02	.020/.020
1	1	4371776	4371767	0.03	0.03	.030/.030
1	1	4371775	4371642	0.04	0.04	.040/.040
4024775	4009233	4371773	4333989	0.06	0	.060/0
4024774	4009232	4371772	4333988	0.06	0.02	.060/.020
4024768	4009221	2	2	0.02	0	.020/0
4024769	4009222	2	2	0.04	0	.040/0
4024771	4009228	2	2	0.08	0	.080/0
4024772	4009230	2	2	0.1	0	.100/0
4024776	4009234	2	2	0.08	0.06	.080/.060

Table 2, Individual Cylinder Liners and Corresponding Kits*						
Existing Cylinder Liner Kit Part Number	Existing Cylinder Liner Part Number	New Cylinder Liner Kit Part Number	New Cylinder Liner Part Number	Upper Press Fit Oversize [inches]	Lower Press Fit Oversize [inches]	Size [inches]
*All product types as included in product affected section above except the following: QSK19 CM850 MCRS, QSK19 CM2150 MCRS and QSK19 CM2350 K105						

Table 3, QSK19 CM850 MCRS and QSK19 CM2150 MCRS Individual Cylinder Liners and Corresponding Kits						
Obsolete Cylinder Liner Kit Part Number	Obsolete Cylinder Liner Part Number	New Cylinder Liner Kit Part Number	New Cylinder Liner Part Number	Upper Press Fit Oversize [inches]	Lower Press Fit Oversize [inches]	Size [inches]
4090054	4916451	4371779	4333991	0	0	0/0
4955809	4927541	4371780	4333993	0.01	0.01	.010/.010
4955795	4927545	4371783	4333996	0.02	0.02	.020/.020
4955810	4927542	4371781	4333994	0.03	0.03	.030/.030
4955811	4927543	4371782	4333995	0.04	0.04	.040/.040
4955808	4927540	2	2	0.02	0	.020/0

Table 4, QSK19 CM2350 K105 Individual Cylinder Liners and Corresponding Kits						
Existing Cylinder Liner Kit Part Number	Existing Cylinder Liner Part Number	New Cylinder Liner Kit Part Number	New Cylinder Liner Part Number	Upper Press Fit Oversize [inches]	Lower Press Fit Oversize [inches]	Size [inches]
4352259	2890916	4371784	4334063	0	0	0/0
Not Applicable	4322664	3	4334065	0.01	0.01	.010/.010
	4322667	3	4334068	0.02	0.02	.020/.020
	4322665	3	4334066	0.03	0.03	.030/.030
	4322666	3	4334067	0.04	0.04	.040/.040

Part Compatibility

⚠ CAUTION ⚠

New crevice seals are compatible with new cylinder liners only. Existing crevice seals are compatible with existing cylinder liners only.

Existing and new cylinder liners can be mixed in the same engine. The new cylinder liners are compatible with all existing power cylinder components.

Part Inventory Action

Use obsolete part number inventory, if available, before using the new part number.

Production Status

The new standard size cylinder liner and new crevice seal has been implemented into production on all product types as included in products affected section above, except the following two product types:

- QSK19 CM850 MCRS
- QSK19 CM2150 MCRS
- QSK19 CM2350 K105

Production information will be updated for the two remaining product types once the existing stock of the standard size cylinder liner is depleted.

Table 5, Production Information

Product Affected	Engine Serial Number (ESN) First	Build Date*	Plant
G19	37266084	30 July 2015	Seymour Engine Plant
K19			
KTA19GC CM558/CM2358 SLB			
QSK19 CM500			
QSK19 G CM558/CM2358			
QSK19 G CM558/CM2358 and CM700/CM2358			
QSK19 CM850 MCRS and QSK19 CM2150 MCRS	Not Available	Not Available	Seymour Engine Plant
QSK19 CM2350 K105	Not Available	Not Available	Seymour Engine Plant

*Engine build date can be found on the engine dataplate.

Publications Affected

Cylinder liner size charts have been updated. See Table 5 for affected publications.

Table 6, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	K19 and G19	4021499	Cylinder Liner	Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html)	1
	QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS	4021592		Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)	
	QSK19 CM2350 K105	4326157		Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/223/223-001-028-tr.html)	
	KTA19GC CM558	4021653		Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/47/47-001-028-tr-gc.html)	
	QSK19G	4021372		Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/48/48-001-028-tr-pwgn.html)	

Document History

Date	Details
2015-2-27	Module Created
2015-8-26	Added QSK19 CM500 to Product Affected section. Update Production Status section and added Table 5.
2015-9-2	Product Problem Solving (PPS)

Last Modified: 02-Sep-2015
